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(54) **HEAT EXCHANGER SYSTEMS AND DEVICES FOR REDUCED PRESSURE LOSS**

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Primary Examiner — Davis D Hwu

(74) *Attorney, Agent, or Firm* — SNELL & WILMER
L.L.P.

(57) **ABSTRACT**

A heat exchanger can comprise: a housing; a heat transfer region disposed within the housing; and an inlet manifold or an inlet conduit defined at least partially by an interior surface extending axially from a first end of the heat exchanger to an inlet defined by the heat transfer region, the inlet manifold or the inlet conduit comprising a dimple configuration disposed in the interior surface. The heat exchanger can further comprise an outlet manifold or an outlet conduit including a dimple configuration disposed on an interior surface of the outlet manifold or outlet conduit.

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(71) Applicant: **Raytheon Technologies Corporation**,
Farmington, CT (US)

(72) Inventors: **Orin G. Avidane**, Lighthouse Point, FL
(US); **Javier A. Cue**, Palm Beach
Gardens, FL (US); **Ryan William
Hunter**, North Palm Beach, FL (US)

(73) Assignee: **RTX Corporation**, Farmington, CT
(US)

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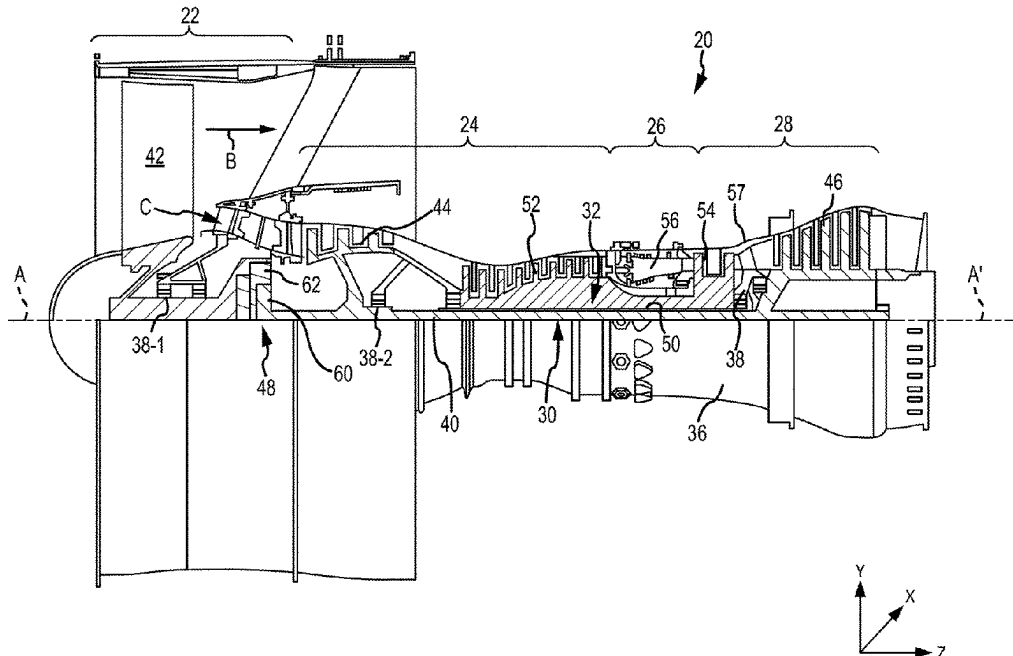
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F28D 9/00 (2006.01)
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(2013.01); **F28D 9/0056** (2013.01)

(58) **Field of Classification Search**
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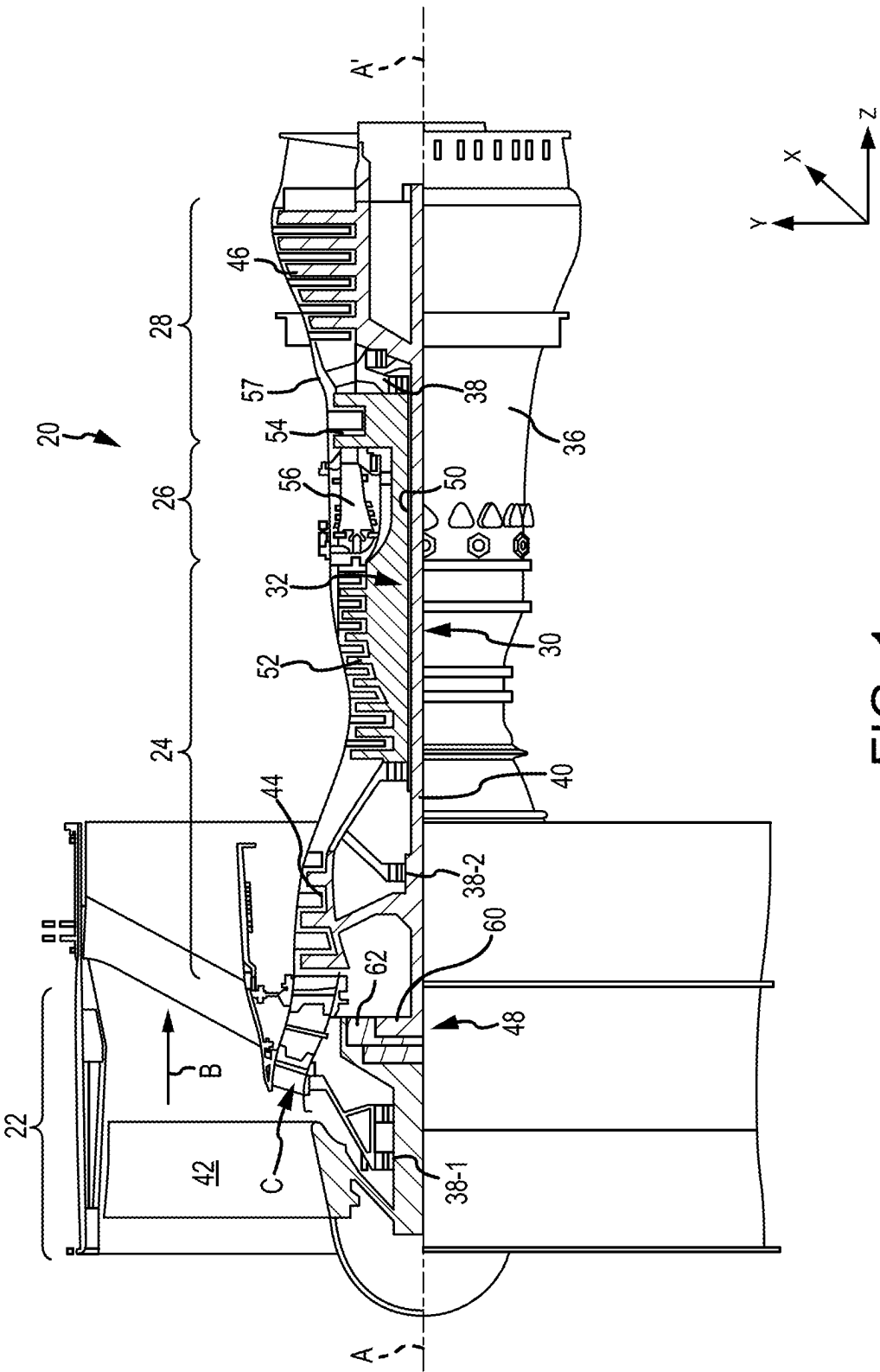


FIG. 1

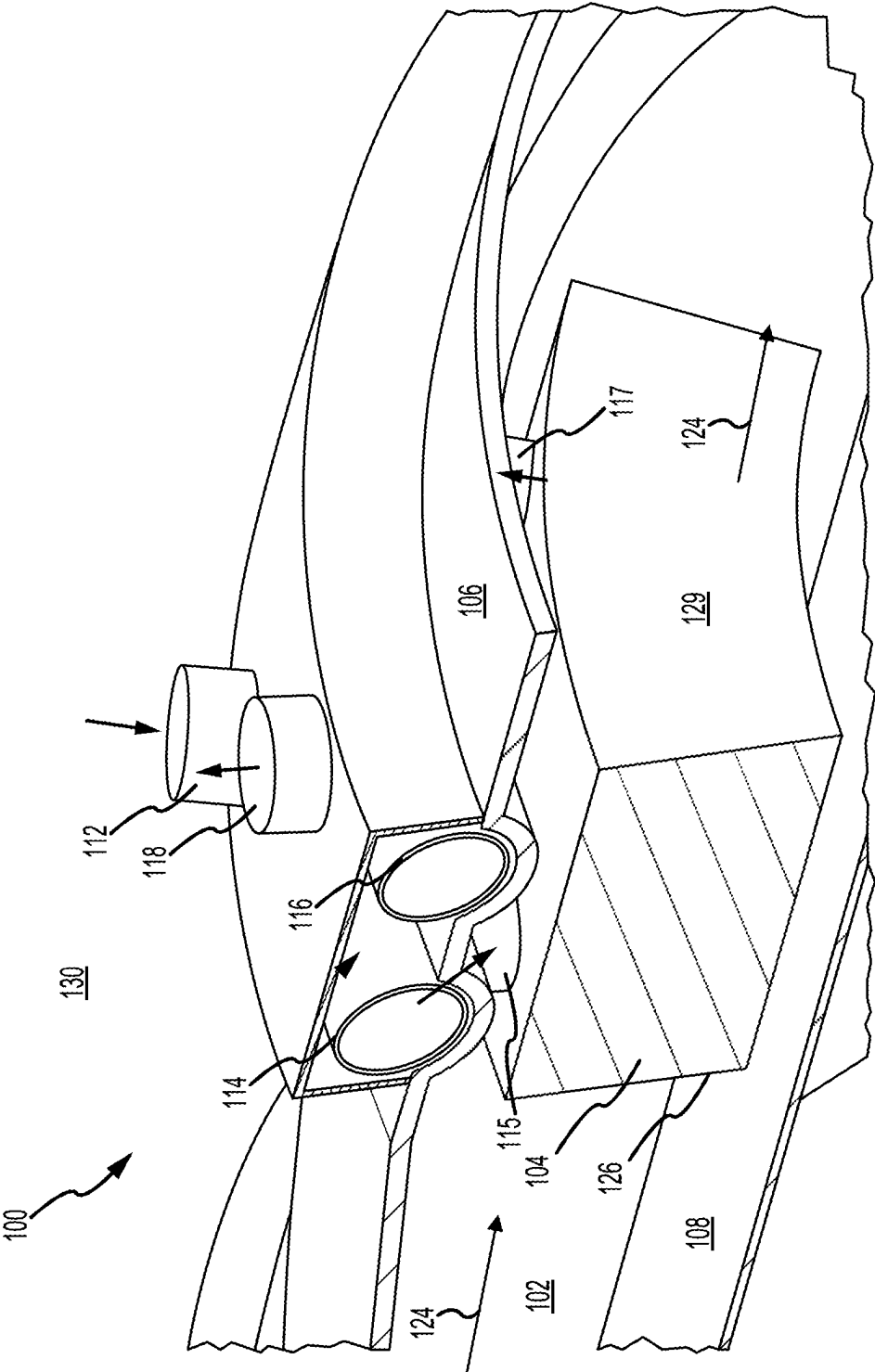


FIG.2

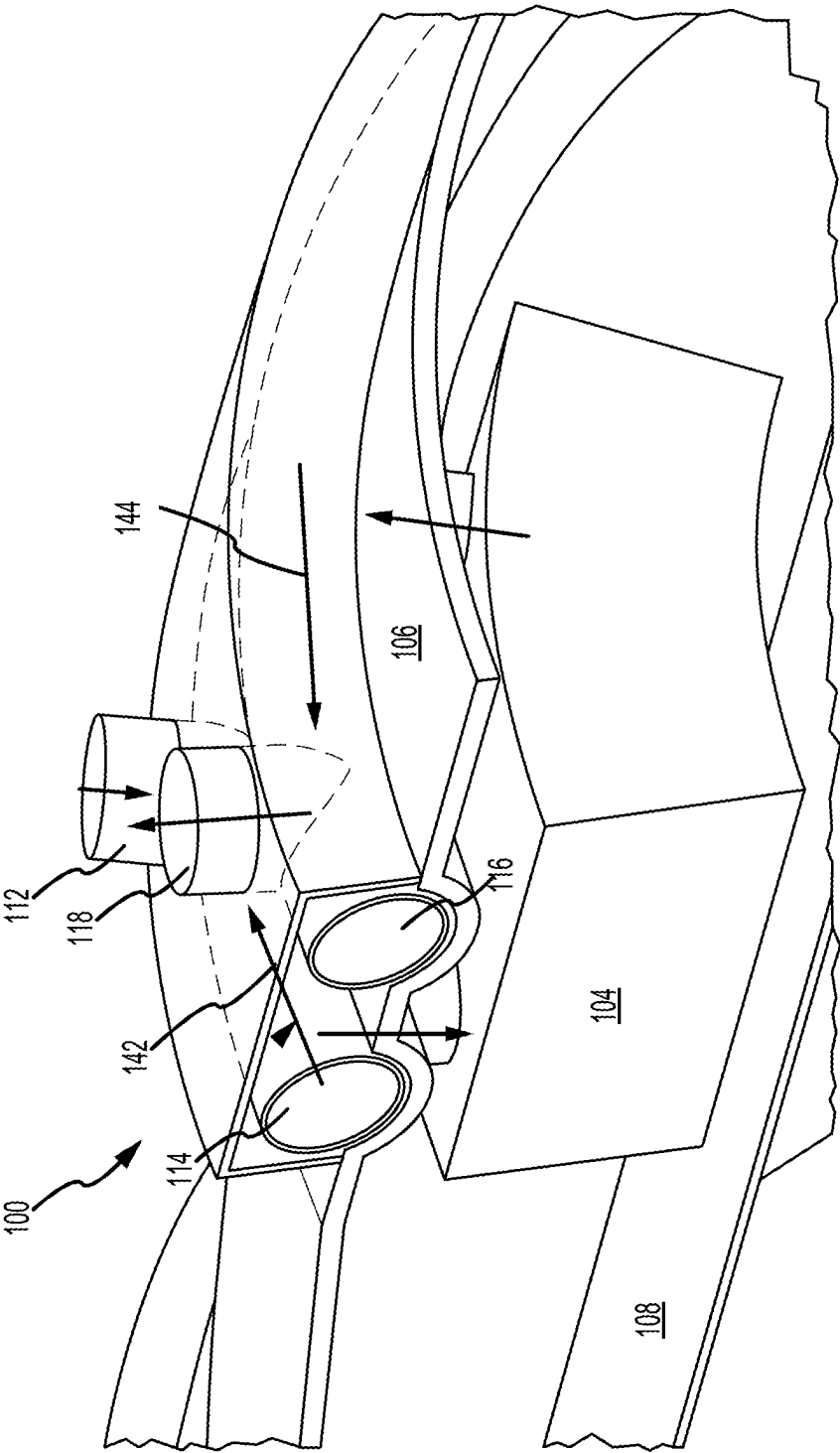


FIG.3

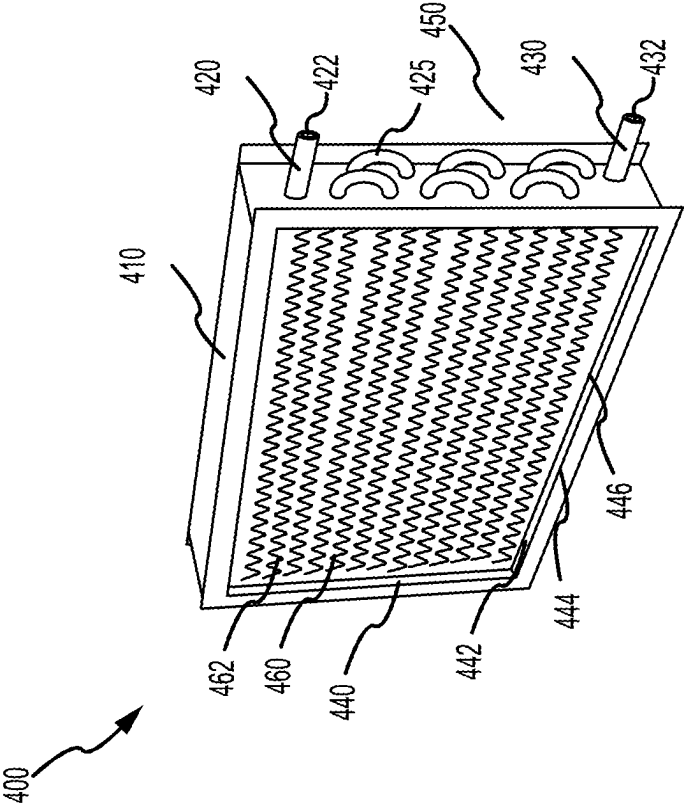


FIG. 4A

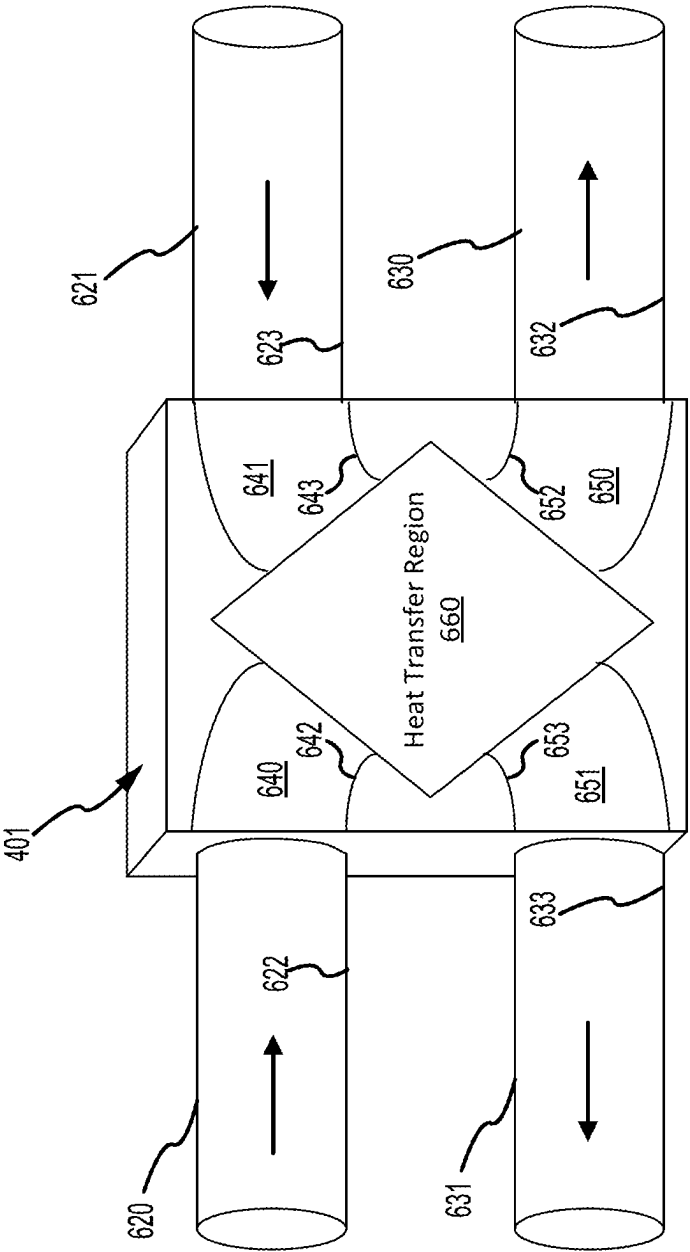


FIG. 4B

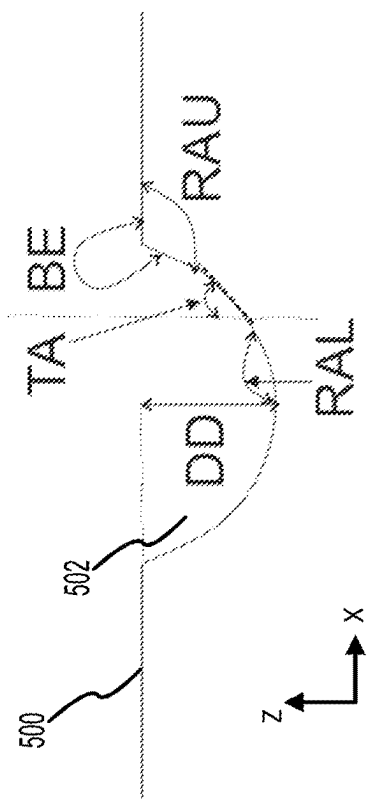


FIG. 5B



FIG. 5C

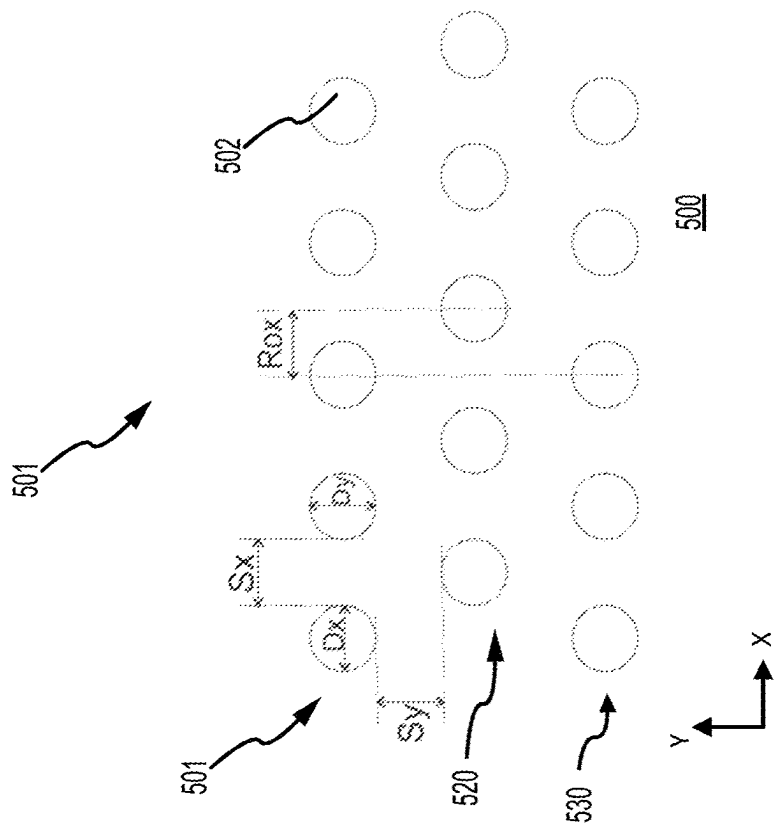


FIG. 5A

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HEAT EXCHANGER SYSTEMS AND DEVICES FOR REDUCED PRESSURE LOSS

FIELD

The present disclosure relates to heat exchangers, and, more specifically, to heat exchanger systems and devices for reduced pressure loss.

BACKGROUND

A gas turbine engine typically includes a fan section, a compressor section, a combustor section, and a turbine section. In general, during operation, air is pressurized in the fan and compressor sections, and is mixed with fuel, and burned in the combustor section to generate hot combustion gases. The hot combustion gases flow through the turbine section, which extracts energy from the hot combustion gases to power the compressor section and other gas turbine engine loads.

Gas turbine engines may include various cooling systems that employ heat exchangers. Bypass air may be directed across the heat exchangers as a cooling fluid. The inlet piping, or outlet piping of the heat exchanger (i.e., the bypass air exiting the heat exchanger) may cause pressure loss that may otherwise be used to drive heat rejection in the heat exchanger. A significant loss in pressure at inlets and outlets of heat exchangers can greatly reduce a heat exchanger performance.

SUMMARY

A heat exchanger is disclosed herein. In various embodiments, the heat exchanger comprises a housing; a heat transfer region disposed within the housing; and an inlet manifold defined at least partially by an interior surface extending axially from a first end of the heat exchanger to an inlet defined by the heat transfer region, the inlet manifold comprising a dimple configuration disposed in the interior surface.

In various embodiments, the dimple configuration comprises a plurality of dimples spaced apart axially along the interior surface. The dimple configuration can further comprise rows of the plurality of dimples spaced apart laterally along the interior surface.

In various embodiments, each dimple in the plurality of dimples comprises a first diameter in a X-direction, a second diameter in a Y-direction, and a depth of dimple, the depth of dimple measured from the interior surface to a max depth of the dimple.

In various embodiments, the heat exchanger further comprises an outlet manifold in fluid communication with the inlet manifold through the heat transfer region, the outlet manifold comprising the dimple configuration. In various embodiments, the heat exchanger further comprises an inlet fluid conduit and an outlet fluid conduit, the inlet fluid conduit in fluid communication with the outlet fluid conduit through the heat transfer region. In various embodiments, the inlet fluid conduit and the outlet fluid conduit each comprise the heat transfer region.

A heat exchanger is disclosed herein. In various embodiments, the heat exchanger comprises a housing; a heat transfer region disposed within the housing; and an inlet fluid conduit extending from a first end to an inlet of the housing, the inlet defined by the heat transfer region, the inlet fluid conduit defining an interior surface and comprising a dimple configuration disposed in the interior surface.

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In various embodiments, the interior surface is a cylindrical surface. In various embodiments, the dimple configuration comprises a plurality of dimples spaced apart axially along the interior surface. In various embodiments, the dimple configuration comprises rows of the plurality of dimples spaced apart circumferentially along the cylindrical surface. In various embodiments, each dimple in the plurality of dimples comprises a first diameter in a X-direction, a second diameter in a Y-direction, and a depth of dimple, the depth of dimple measured from the interior surface to a max depth of the dimple.

In various embodiments, the heat exchanger further comprises an outlet fluid conduit in fluid communication with the inlet fluid conduit through a fluid conduit in the heat transfer region, the outlet fluid conduit comprising the dimple configuration. In various embodiments, the fluid conduit in the heat transfer region is without dimples.

In various embodiments, the heat exchanger further comprises an inlet manifold and an outlet manifold, the inlet manifold in fluid communication with the outlet manifold through the heat transfer region.

A gas turbine engine is disclosed herein. In various embodiments, the gas turbine engine comprises: a heat exchanger comprising a heat transfer region; an inlet manifold and an outlet manifold; the outlet manifold in fluid communication with the inlet manifold through the heat transfer region; a supply conduit; and a return conduit in fluid communication with the supply conduit through the heat transfer region of the heat exchanger, wherein an interior surface of at least one of the supply conduit, the return conduit, the inlet manifold and the outlet manifold comprises a dimple configuration.

The foregoing features and elements may be combined in various combinations without exclusivity, unless expressly indicated otherwise. These features and elements as well as the operation thereof will become more apparent in light of the following description and the accompanying drawings. It should be understood, however, the following description and drawings are intended to be exemplary in nature and non-limiting.

BRIEF DESCRIPTION OF THE DRAWINGS

The subject matter of the present disclosure is particularly pointed out and distinctly claimed in the concluding portion of the specification. A more complete understanding of the present disclosure, however, may best be obtained by referring to the detailed description and claims when considered in connection with the figures, wherein like numerals denote like elements.

FIG. 1 illustrates a cross-sectional view of an exemplary gas turbine engine, in accordance with various embodiments;

FIG. 2 illustrates a perspective view of a cooling system having a plenum, in accordance with various embodiments;

FIG. 3 illustrates a perspective view of a cooling system having a plenum, in accordance with various embodiments;

FIG. 4A illustrates a perspective view of a heat exchanger, in accordance with various embodiments;

FIG. 4B illustrates a side view of a heat exchanger, in accordance with various embodiments;

FIG. 5A illustrates a top view of an interior surface with a dimple configuration, in accordance with various embodiments; and

FIG. 5B illustrates a cross-sectional view of the interior surface with a dimple of the dimple configuration, in accordance with various embodiments.

FIG. 5C illustrates an elliptical view of the interior surface with a dimple of the dimple configuration, in accordance with various embodiments.

DETAILED DESCRIPTION

The detailed description of exemplary embodiments herein makes reference to the accompanying drawings, which show exemplary embodiments by way of illustration. While these exemplary embodiments are described in sufficient detail to enable those skilled in the art to practice the exemplary embodiments of the disclosure, it should be understood that other embodiments may be realized and that logical changes and adaptations in design and construction may be made in accordance with this disclosure and the teachings herein. Thus, the detailed description herein is presented for purposes of illustration only and not limitation. The steps recited in any of the method or process descriptions may be executed in any order and are not necessarily limited to the order presented.

Furthermore, any reference to singular includes plural embodiments, and any reference to more than one component or step may include a singular embodiment or step. Also, any reference to attached, fixed, connected or the like may include permanent, removable, temporary, partial, full and/or any other possible attachment option. Additionally, any reference to without contact (or similar phrases) may also include reduced contact or minimal contact. Surface cross hatching lines may be used throughout the figures to denote different parts, but not necessarily to denote the same or different materials.

Throughout the present disclosure, like reference numbers denote like elements. Accordingly, elements with like element numbering may be shown in the figures, but may not necessarily be repeated herein for the sake of clarity.

As used herein, “aft” refers to the direction associated with the tail (e.g., the back end) of an aircraft, or generally, to the direction of exhaust of a gas turbine engine.

A first component that is “radially outward” of a second component means that the first component is positioned at a greater distance away from a central longitudinal axis of the gas turbine engine. A first component that is “radially inward” of a second component means that the first component is positioned closer to the engine central longitudinal axis than the second component. The terminology “radially outward” and “radially inward” may also be used relative to reference axes other than the engine central longitudinal axis.

Disclosed herein are systems and apparatuses for improving a heat exchanger performance. Although disclosed herein with reference to gas turbine engines, the present disclosure is not limited in this regard. For example, any heat exchanger used in any type of application is within the scope of this disclosure. In various embodiments, an interior surface of an inlet manifold and/or an outlet manifold for a heat exchanger comprises a dimpled configuration. In various embodiments, the inlet manifold and/or the outlet manifold can be integral with the heat exchanger. In various embodiments, the inlet manifold and/or the outlet manifold can be configured to be coupled to the heat exchanger. The present disclosure is not limited in this regard. In various embodiments, the dimpled configuration may result in less separation of a boundary layer against the interior surface. In this regard, the dimpled configuration can reduce a pressure loss at an inlet and/or an outlet of a heat exchanger, resulting in improved performance of the heat exchanger, in accordance with various embodiments.

In various embodiments and with reference to FIG. 1, a gas turbine engine 20 is provided. Gas turbine engine 20 may generally include a fan section 22, a compressor section 24, a combustor section 26, and a turbine section 28. In operation, fan section 22 drives fluid (e.g., air) along a bypass flow-path B, while compressor section 24 drives fluid along a core flow-path C for compression and communication into combustor section 26 and then expansion through turbine section 28. Although gas turbine engine 20 is depicted as a turbofan gas turbine engine herein, it should be understood that the concepts described herein are not limited to use with turbofans as the teachings may be applied to other types of turbine engines.

Gas turbine engine 20 may generally comprise a low speed spool 30 and a high speed spool 32 mounted concentrically, via bearing systems 38, for rotation about for rotation about engine central longitudinal axis A-A' and relative to an engine static structure 36. It should be understood that various bearing systems 38 at various locations may alternatively or additionally be provided, including for example, bearing system 38, bearing system 38-1, and bearing system 38-2. Engine central longitudinal axis A-A' is oriented in the z direction on the provided xyz axes. The z direction on the provided xyz axes refers to the axial direction. As used herein, the term “radially” refer to directions towards and away from engine central longitudinal axis A-A' and the z-axis. As used herein, the terms “circumferential” and “circumferentially” refer to directions about central longitudinal axis A-A' and the z-axis.

Low speed spool 30 may generally comprise an inner shaft 40 that interconnects a fan 42, a low-pressure compressor 44, and a low-pressure turbine 46. Inner shaft 40 may be connected to fan 42 through a geared architecture 48 that can drive fan 42 at a lower speed than low speed spool 30. Geared architecture 48 may comprise a gear assembly 60 enclosed within a gear housing 62. Gear assembly 60 couples inner shaft 40 to a rotating fan structure. High speed spool 32 may comprise an outer shaft 50 that interconnects a high-pressure compressor 52 and high-pressure turbine 54. A combustor 56 may be located between high pressure compressor 52 and high-pressure turbine 54. A mid-turbine frame 57 of engine static structure 36 may be located generally between high pressure turbine 54 and low-pressure turbine 46. Mid-turbine frame 57 may support one or more bearing systems 38 in turbine section 28. Inner shaft 40 and outer shaft 50 may be concentric and rotate via bearing systems 38 about engine central longitudinal axis A-A', which is collinear with their longitudinal axes. As used herein, a “high pressure” compressor or turbine experiences a higher pressure than a corresponding “low pressure” compressor or turbine. The airflow in core flow-path C may be compressed by low pressure compressor 44 and high-pressure compressor 52, mixed and burned with fuel in combustor 56, then expanded over high-pressure turbine 54 and low-pressure turbine 46. Turbines 46, 54 rotationally drive the respective low speed spool 30 and high-speed spool 32 in response to the expansion.

The compressor section 24, the combustor section 26, and the turbine section 28 are generally referred to as the engine core. Air is drawn into gas turbine engine 20 through fan 42. Air exiting fan 42 may be divided between core flow-path C and bypass flow-path B. The airflow in bypass flow-path B may be utilized for multiple purposes including, for example, cooling and pressurization.

Referring to FIG. 2, and with continued reference to FIG. 1, a cooling system 100 having a heat exchanger 104 is illustrated, in accordance with various embodiments. The

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heat exchanger **104** may be located between an outer engine case structure **106** and an inner engine case structure **108**. Outer engine case structure **106** is radially outward of inner engine case structure **108**. Outer engine case structure **106** and inner engine case structure **108** may define a generally annular bypass duct **102** around the engine core. In various embodiments, inner engine case structure **108** may form a portion of engine static structure **36**. In various embodiments, air discharged from, for example, fan section **22** may be communicated through the bypass duct **102**.

In various embodiments, the cooling system **100** comprise a supply conduit **114** and a return conduit **116**. In accordance with various embodiments, an inlet conduit **112** may be fluidly coupled between an external source from an aircraft proximate engine bay **130** and a supply conduit **114**. Engine bay **130** may be disposed radially outward from outer engine case structure **106**. Supply conduit **114** may extend circumferentially about outer engine case structure **106** and/or form an annular tube. The external source from the aircraft may be disposed radially outward of outer engine case structure **106**. Inlet conduit **112** directs hot air (i.e., air to be cooled) to heat exchanger **104**. Inlet conduit **112** is fluidly coupled to an inlet conduit **115** of heat exchanger **104**. Supply conduit **114** may be fluidly coupled to each heat exchanger **104** by the inlet conduit **115** extending radially inward from supply conduit **114** through outer engine case structure **106**.

In accordance with various embodiments, an outlet conduit **118** may be fluidly coupled between an external destination from an aircraft proximate engine bay **130** and return conduit **116**. Return conduit **116** may extend circumferentially about outer engine case structure **106** and/or form an annular tube. The external destination from the aircraft may be disposed radially outward of outer engine case structure **106**. Outlet conduit **118** directs cool air (i.e., air that is cooled by heat exchanger **104**) to the external destination. Outlet conduit **118** is fluidly coupled to an outlet conduit **117** of heat exchanger **104**. Return conduit **116** may be fluidly coupled to each heat exchanger **104** by the outlet conduit **117** extending radially inward from return conduit **116** through outer engine case structure **106**.

The hot air provided by inlet conduit **112** is cooled in heat exchanger **104** and then returned to the external source from the aircraft through a return conduit **116**. Return conduit **116** is fluidly coupled to an outlet conduit **117** of heat exchanger **104** and outlet conduit **118**. The cooled air in return conduit **116** may be directed downstream to cool components in thermally challenged regions.

Heat exchanger **104** is configured to receive a cooling airflow, for example, airflow **124**. In various embodiments, airflow **124** may be a portion of the bypass airflow in bypass duct **102**. Cooling airflow **124** may be directed through an inlet manifold **126** across heat exchanger **104** and out an outlet manifold **129** to cool the air supplied from inlet conduit **115** in heat exchanger **104**.

Referring to FIG. 3, a cooling system **100** having a heat exchanger **104** is illustrated, in accordance with various embodiments. In various embodiments, inlet conduit **112** may be integral to supply conduit **114**. Similarly, outlet conduit **118** may be integral to return conduit **116**. "Integral," as referred to herein is a monolithic component. In various embodiments, hot air may flow through supply conduit **114** in a first circumferential direction **142** and the return cooler air may flow through return conduit **116** in a second circumferential direction **144**. The first circumferential direction **142** and the second circumferential direction **144** may be opposite directions.

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Referring now to FIG. 4, a perspective view of a heat exchanger **400** is illustrated, in accordance with various embodiments. In various embodiments, the heat exchanger **104** from FIGS. 2 and 3 can be in accordance (i.e., can contain the same features), with the heat exchanger **400** as described further herein. In various embodiments, the heat exchanger **400** comprises a housing **410**, an inlet conduit **420**, an outlet conduit **430**, an inlet manifold **440**, an outlet manifold **450**, and a heat transfer region **460**. "A heat transfer region" as referred to herein is a portion of the heat exchanger where a first fluid (e.g., air, a working fluid or the like) interacts with a second fluid (e.g., air, a working fluid or the like), to transfer heat from the first fluid to the second fluid or vice versa. In various embodiments, the first fluid and the second fluid can be any fluid for use in a heat exchanger. For example, air, water, fuel, oil, or any other fluid utilized in heat exchangers are within the scope of this disclosure.

In various embodiments, the inlet conduit **420** is in fluid communication with the outlet conduit **430** through the heat transfer region **460**. For example, a fluid conduit **425** extends through the heat transfer region **460**, and sometimes out of the heat transfer region **460** from the inlet conduit **420** to the outlet conduit **430**. Similarly, the inlet manifold **440** is in fluid communication with the outlet manifold **450** through the heat transfer region **460**.

In various embodiments, the inlet manifold **440** is defined by an interior surface **442** that extends axially from a first edge **444** disposed at a first axial end of the heat exchanger **400** to a second edge **446** defined by an inlet **462** to the heat transfer region **460**. Similarly, in accordance with various embodiments, the outlet manifold **450** is in accordance with the inlet manifold (i.e., with an interior surface that extends axially from a first edge defined by an outlet of the heat transfer region to a second edge defined by a second axial end of the heat exchanger **400**).

In various embodiments, the inlet conduit **420** comprises an interior surface **422** and the outlet conduit **430** comprises an interior surface **432**. Although described herein as having an inlet manifold **440** and an outlet manifold **450** with an inlet conduit **420** and an outlet conduit **430**, the present disclosure is not limited in this regard. For example, a heat exchanger with two inlet manifolds and two outlet manifolds or a heat exchanger with two inlet conduits and two outlet conduits (e.g., FIG. 4B) are within the scope of this disclosure.

Referring now to FIG. 4B, a side view of a heat exchanger **401** with two inlet conduits (e.g., inlet conduit **620** and inlet conduit **621**) and two outlet conduits (e.g., outlet conduit **630** and outlet conduit **631**) is illustrated, in accordance with various embodiments. In various embodiments, the inlet conduits **620**, **621** are in accordance with the inlet conduit **420**, and the outlet conduits **630**, **631** are in accordance with the outlet conduit **430** from FIG. 4A.

In various embodiments, the heat exchanger **401** comprises inlet manifolds **640**, **641** and outlet manifolds **650**, **651**. For example, the inlet manifold **640** can be disposed between the inlet conduit **620** and the heat transfer region **660** within the heat exchanger **401**. Similarly, the inlet manifold **641** can be disposed between the inlet conduit **621** and the heat transfer region **660**, the outlet manifold **650** can be disposed between the outlet conduit **630** and the heat transfer region **660**, and the outlet manifold **651** can be disposed between the outlet conduit **631**. However, the present disclosure is not limited in this regard. For example,

an outlet of the inlet conduit **620** can be the heat transfer region **660** directly and would still be within the scope of this disclosure.

In various embodiments, a first fluid is supplied through the inlet conduit **620** to a heat transfer region **660** and exhausted out the outlet conduit **630**. Similarly, a second fluid is supplied through the inlet conduit **621** to the heat transfer region **660** and out the outlet conduit **631**. In various embodiments, the first fluid is at a first temperature that is greater than the second fluid. In this regard, the heat transfer region **660** is configured to decrease the first temperature of the first fluid and increase the temperature of the second fluid. For example, in accordance with various embodiments, an air-to-air heat exchanger may extract the first fluid that is 22 degree F. above ambient in the inlet conduit **620** and extract a second fluid that is 10 degrees F. below ambient from the inlet conduit **621**. As a result, in accordance with various embodiments, the second fluid may be heated up to approximately 15 degrees F. above ambient and exhausted out the outlet conduit **631**, and the first fluid may be cooled down to approximately 5 degrees above ambient and exhausted out the outlet conduit **630**.

Although described with respect to an air-to-air heat exchanger, the present disclosure is not limited in this regard. Any fluid-to-fluid heat exchanger is within the scope of this disclosure. For example, various fluids and media may be circulated throughout the heat exchangers **400**, **401** during operation and may carry heat including, without limitation, air, fuel, oil, lubricating fluid, hydraulic fluid, thermally neutral heat-transfer fluid, or any other fluid suitable for circulating through a heat exchanger **400**, **401**.

In various embodiments, each inlet conduit, outlet conduit, inlet manifold and outlet manifold of the heat exchanger **401** each comprise an interior surface configured to reduce a pressure loss in the heat exchanger **401**. For example, the inlet conduit **620** comprises an interior surface **622**, the inlet conduit **621** comprises an interior surface **623**, the outlet conduit **630** comprises an interior surface **632**, the outlet conduit **631** comprises an interior surface **633**, the inlet manifold **640** comprises an interior surface **642**, the inlet manifold **641** comprises an interior surface **643**, the outlet manifold **650** comprises an interior surface **652**, and the outlet manifold **651** comprises an interior surface **653**.

Referring now to FIGS. **5A** and **5B**, a top view (FIG. **5A**) and a cross-sectional view (FIG. **5B**) of an interior surface **500** (e.g., any of interior surfaces **422**, **432**, **442**, **622**, **623**, **632**, **633**, **642**, **643**, **652**, **653** and/or an interior surface of outlet manifold **450** from FIGS. **4A** and **4B**) is illustrated, in accordance with various embodiments. In various embodiments, the interior surface can be a cylindrical surface (e.g., for inlet conduit **420**, **620**, **621** and/or outlet conduit **430**, **630**, **631**), a flat surface (e.g., for inlet manifold **440** or outlet manifold **450**), or the like. The present disclosure is not limited in this regard.

In various embodiments, the interior surface **500** comprises a dimple configuration **501** with a plurality of dimples (e.g., dimple **502**). In various embodiments, each dimple (e.g., dimple **502**) can be the same for ease of manufacturing. However, the present disclosure is not limited in this regard, and varying dimple sizes (i.e., diameter and depth) throughout the dimple configuration **501** is within the scope of this disclosure. In various embodiments, the dimple **502** comprises a diameter in an X-direction (“Dx”) and a diameter in a Y-direction (“Dy”). The X-direction can correspond to an axial direction (i.e., an axial direction of a fluid conduit or an axial direction of a manifold as shown in FIG. **4**), and the Y-direction can correspond to a circumferential direction

(i.e., for a fluid conduit) or a lateral direction (i.e., for a manifold defined by a flat surface). From a top view (FIG. **5A**), the dimple **502** can comprise an elliptical shape (e.g., as shown in FIG. **5C**), in accordance with various embodiments. Although the dimple **502** is illustrated as a circular cross-section in an XY plane, the present disclosure is not limited in this regard. For example, diameter Dx can vary between 0.007 inches (0.018 cm) and 2.9 inches (7.366 cm), and diameter Dy can vary between 0.007 inches (0.018 cm) and 2.9 inches (7.366 cm), in accordance with various embodiments.

In various embodiments, dimples **502** can be spaced apart in the X-direction (“Sx”), and the dimples **502** can be spaced apart in the Y-direction (“Sy”). For example, Sy and Sx can each vary between 0.001 inches (0.0025 cm) and 1.9 inches (4.83 cm). Although illustrated as being spaced apart equally in the x-direction and spaced apart equally in the y-direction, the present disclosure is not limited in this regard. For example, the dimple configuration **501** can comprise arrays of dimples (e.g., two or more dimples **502** spaced close together (e.g., 0.001 inches (0.0025 cm) then a larger gap (e.g., 1 inch (2.54 cm)), then have the pattern repeat.

In various embodiments, the dimple configuration **501** comprises a plurality of dimples arranged in rows (e.g., rows **510**, **520**, **530** as shown in FIG. **5A**). In various embodiments, adjacent rows (e.g., rows **510**, **520**) can be offset in an X direction (“Rox”). In various embodiments, the Rox corresponds to two times the Sx direction.

Referring now to FIG. **5B**, a cross-sectional view of a dimple **502** from FIG. **5A** is illustrated, in accordance with various embodiments. In various embodiments, the dimple **502** further comprise a ramp angle upper (“RAU”), a ramp angle lower (“RAL”), a transition angle (“TA”), a break angle (“BE”), and a depth of dimple (“DD”). The RAU is an angle between the interior surface **500**, and an interior surface of the dimple **502**. The TA is an angle from a vertical reference (i.e., a normal direction from the interior surface **500**), and a tangent line between a vertex defined by the dimple **502** and the interior surface **500** and a point defined by the DD. The RAL is an angle between the tangent line and a horizontal plane defined by the DD. The break edge is an angle that transitions the interior surface **500** to an interior surface of the dimple **502**.

In various embodiments, the dimple **502** can be a partial sphere. In various embodiments, the dimple **502** can have a depth greater than diameter Dx and diameter Dy or less than diameter Dx and Dy, the present disclosure is not limited in this regard. In various embodiments, the dimple **502** is configured to induce turbulent flow. In various embodiments, turbulent flow can result in reduced pressure loss relative to laminar flow.

In various embodiments, the RAU can be between 90 degrees and 179 degrees. In various embodiments, the RAL can be between 0 degrees and 90 degrees. In various embodiments, the TA can be between 0 degrees and 89 degrees. In various embodiments, the BE can be between 181 degrees and 270 degrees. In various embodiments, the DD can be between 0.001 inches (0.0025 cm) and 1.5 inches (3.81 cm).

In various embodiments, the interior surface **500** can be substantially an entire length of an inlet conduit (e.g., inlet conduit **420** from FIG. **4**), an outlet conduit (e.g., outlet conduit **430**), an inlet manifold (e.g., inlet manifold **440**), an outlet manifold (e.g., outlet manifold **450**), a supply conduit (e.g., supply conduit **114**), a return conduit (e.g., return conduit **116**), or the like. The present disclosure is not limited in this regard. In various embodiments, “substan-

tially the entire length” as referred to herein is between 95% and 100% of a total length as measured from a first end of a conduit (or manifold) to a second end of a conduit (or manifold), the second end being proximate (or adjacent) the heat transfer region **460** of a heat exchanger **400** from FIG. **4**, in accordance with various embodiments.

In various embodiments, by having a dimple configuration **501** as disclosed herein, a pressure loss of a fluid entering and exiting a heat exchanger can be reduced, thereby increasing an efficiency of the heat exchanger (e.g., heat exchanger **104** or heat exchanger **400**).

Benefits, other advantages, and solutions to problems have been described herein with regard to specific embodiments. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical system. However, the benefits, advantages, solutions to problems, and any elements that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as critical, required, or essential features or elements of the disclosures. The scope of the disclosures is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” Moreover, where a phrase similar to “at least one of A, B, or C” is used in the claims, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C. Different cross-hatching is used throughout the figures to denote different parts but not necessarily to denote the same or different materials.

Systems, methods and apparatus are provided herein. In the detailed description herein, references to “one embodiment”, “an embodiment”, “an example embodiment”, etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments.

Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element is intended to invoke 35 U.S.C. 112(f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprises”, “comprising”, or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

What is claimed is:

1. A heat exchanger, comprising:

a housing;

a heat transfer region disposed within the housing; and
an inlet manifold defined at least partially by an interior surface recessed from a first edge to a second edge, the second edge extending axially from a first end of the heat exchanger to an inlet defined by the heat transfer region and disposed transversely with respect to the inlet manifold, the inlet manifold comprising a dimple configuration disposed in the interior surface.

2. The heat exchanger of claim **1**, wherein the dimple configuration comprises a plurality of dimples spaced apart axially along the interior surface.

3. The heat exchanger of claim **2**, wherein the dimple configuration comprises rows of the plurality of dimples spaced apart laterally along the interior surface.

4. The heat exchanger of claim **2**, wherein each dimple in the plurality of dimples comprises a first diameter in a X-direction, a second diameter in a Y-direction, and a depth of dimple, the depth of dimple measured from the interior surface to a max depth of the dimple.

5. The heat exchanger of claim **4**, wherein the first diameter in the X-direction and the second diameter in the Y-direction are each between 0.007 inches (0.18 cm) and 2.9 inches (7.4 cm), and the depth of dimple is between 0.001 inches (0.0025 cm) and 1.5 inches (3.81 cm).

6. The heat exchanger of claim **1**, further comprising:

an outlet manifold recessed axially from a second end of the heat exchanger opposing the first end and in fluid communication with the inlet manifold through the heat transfer region, the outlet manifold comprising the dimple configuration; and

an inlet fluid conduit and an outlet fluid conduit, the inlet fluid conduit in fluid communication with the outlet fluid conduit through the heat transfer region.

7. The heat exchanger of claim **6**, wherein the inlet fluid conduit and the outlet fluid conduit each comprise the heat transfer region and extend transversely with respect to the inlet manifold and the outlet manifold.

8. A heat exchanger, comprising:

a housing;

a heat transfer region disposed within the housing;

an inlet fluid conduit extending from a first end to an inlet of the housing, the inlet defined by the heat transfer region, the inlet fluid conduit defining an interior surface and comprising a dimple configuration disposed in the interior surface; and

an inlet manifold and an outlet manifold, the inlet manifold in fluid communication with the outlet manifold through the heat transfer region.

9. The heat exchanger of claim **8**, wherein the interior surface is a cylindrical surface.

10. The heat exchanger of claim **9**, wherein the dimple configuration comprises a plurality of dimples spaced apart axially along the interior surface.

11. The heat exchanger of claim **10**, wherein the dimple configuration comprises rows of the plurality of dimples spaced apart circumferentially along the cylindrical surface.

12. The heat exchanger of claim **10**, wherein each dimple in the plurality of dimples comprises a first diameter in a X-direction, a second diameter in a Y-direction, and a depth of dimple, the depth of dimple measured from the interior surface to a max depth of the dimple.

13. The heat exchanger of claim **4**, wherein the first diameter in the X-direction and the second diameter in the Y-direction are each between 0.007 inches (0.18 cm) and 2.9

inches (7.4 cm), and the depth of dimple is between 0.001 inches (0.0025 cm) and 1.5 inches (3.81 cm).

14. The heat exchanger of claim 8, further comprising an outlet fluid conduit in fluid communication with the inlet fluid conduit through a fluid conduit in the heat transfer region, the outlet fluid conduit comprising the dimple configuration, wherein the fluid conduit in the heat transfer region is without dimples.

15. A gas turbine engine comprising the heat exchanger of claim 1, the gas turbine engine further comprising:
a supply conduit; and
a return conduit in fluid communication with the supply conduit through the heat transfer region of the heat exchanger.

16. A gas turbine engine comprising the heat exchanger of claim 8, the gas turbine engine further comprising:
a supply conduit; and
a return conduit in fluid communication with the supply conduit through the heat transfer region of the heat exchanger.

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